

Economic Freedom Index 2019 — Descriptive Report

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2025-10-14

Introduction to the Economic Freedom Index 2019 data set

Context

The Economic Freedom Index is poised to help readers track over two decades of the advancement in economic freedom, prosperity, and opportunity and promote these ideas in their homes, schools, and communities. The Index covers 12 freedoms – from property rights to financial freedom – in 186 countries. The central variable is 2019 Score, which summarizes the degree of economic freedom with a score from 0-100. A higher score indicates greater economic freedom.

Content

The data (last updated 26/02/2019) is presented in CSV format as follows: CountryID, Country Name, WEBNAME, Region, World Rank, Region Rank, 2019 Score, Property Rights, Judicial Effectiveness, Government Integrity, Tax Burden, Gov't Spending, Fiscal Health, Business Freedom, Labor Freedom, Monetary Freedom, Trade Freedom, Investment Freedom, Financial Freedom, Tariff Rate (%), Income Tax Rate (%), Corporate Tax Rate (%), Tax Burden % of GDP, Gov't Expenditure % of GDP, Country, Population (Millions), "GDP (Billions, PPP)", GDP Growth Rate (%), 5 Year GDP Growth Rate (%), GDP per Capita (PPP), Unemployment (%), Inflation (%), FDI Inflow (Millions), Public Debt (% of GDP).

Goal

The goal of this analysis is to conduct a descriptive study of the Economic Freedom Index 2019 dataset to examine the factors that influence economic freedom across different nations and regions. By exploring the distribution of the 2019 scores and their associations with economic, institutional, and social variables, we aim to understand how economic freedom varies globally. The findings from this analysis can help inform policy decisions that promote higher levels of economic freedom and sustainable economic development worldwide.

Research Questions

The analysis will address the following key inquiries:

- What are the top 5 countries with the highest levels of economic freedom in 2019?
- What are the 5 countries with the lowest levels of economic freedom in 2019?
- How does GDP per capita relate to economic freedom?
- Do countries with stronger property rights tend to exhibit higher economic freedom?
- What is the relationship between government integrity and economic freedom?
- Do countries with greater trade openness experience higher economic freedom?
- Are there regional differences in economic freedom scores across the world?
- What patterns or similarities emerge between countries with high and low economic freedom?

This analysis will provide an in-depth exploration of these questions, aiming to uncover patterns that can help guide policies to strengthen the institutional and economic conditions that foster greater economic freedom globally.

Variable Description

The Economic Freedom Index 2019 dataset contains several key variables that aim to quantify economic freedom and its contributing factors across various countries. Below is a description of the primary variables used in this analysis:

- Country Name: The name of each country included in the Economic Freedom Index.
- Region: The geographical region where the country is located.
- 2019 Score: The overall measure of economic freedom on a scale from 0 to 100, where higher values indicate greater freedom.
- Property Rights: Index measuring the degree to which private property is protected by the law and enforcement is effective.
- Judicial Effectiveness: Reflects the efficiency and fairness of the judicial system, including independence of courts.
- Government Integrity: Indicates the level of transparency and freedom from corruption in government institutions.
- Trade Freedom: Measures the openness of a country's trade policy, based on tariffs and non-tariff barriers.
- GDP per Capita (PPP): Gros Domestic Product per person, adjusted for purchasing power parity, in international dollars.
- GDP Growth Rate (%): The percentage change in a country's GDP from the previous year.
- Population (Millions): Total population of the country, expressed in millions.

This variable description provides a detailed understanding of the dataset, allowing for a clearer interpretation of the analysis results.

Package Installation and Loading

```
# Install packages only if not already installed
if (!require("tidyverse")) install.packages("tidyverse")

## Loading required package: tidyverse

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr    1.5.2
## v ggplot2     4.0.0      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

if (!require("ggplot2")) install.packages("ggplot2")
if (!require("dplyr")) install.packages("dplyr")
if (!require("ggpubr")) install.packages("ggpubr")

## Loading required package: ggpubr
if (!require("reshape2")) install.packages("reshape2")

## Loading required package: reshape2
##
## Attaching package: 'reshape2'
##
## The following object is masked from 'package:tidyr':
##
##      smiths
if (!require("janitor")) install.packages("janitor")

## Loading required package: janitor
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##      chisq.test, fisher.test
# Load libraries
library(tidyverse)
library(ggplot2)
library(dplyr)
library(ggpubr)
library(reshape2)
library(janitor)
library(corrplot)

## corrplot 0.95 loaded
# Read the Economic Freedom Index Data

economic_freedom_data = read.csv("economic_freedom_index2019_data.csv")

```

Descriptive Analysis

```

# Changes data type from character to numeric

economic_freedom_data$X2019.Score <- as.numeric(economic_freedom_data$X2019.Score)
economic_freedom_data$Property.Rights <- as.numeric(economic_freedom_data$Property.Rights)
economic_freedom_data$Judicial.Effectiveness <- as.numeric(economic_freedom_data$Judicial.Effectiveness)
economic_freedom_data$Government.Integrity <- as.numeric(economic_freedom_data$Government.Integrity)
economic_freedom_data$Trade.Freedom <- as.numeric(economic_freedom_data$Trade.Freedom)
economic_freedom_data$GDP.per.Capita..PPP. <- as.numeric(gsub("[$,]", "", economic_freedom_data$GDP.per)
economic_freedom_data$GDP.Growth.Rate.... <- as.numeric(economic_freedom_data$GDP.Growth.Rate....)
economic_freedom_data$Population..Millions. <- as.numeric(economic_freedom_data$Population..Millions.)

```

Summary Statistics

```
# Provides summary statistics for variables

summary_vars <- economic_freedom_data %>%
  select(X2019.Score, Property.Rights, Judicial.Effectiveness, Government.Integrity,
         Trade.Freedom, GDP.per.Capita..PPP., GDP.Growth.Rate..., Population..Millions.)

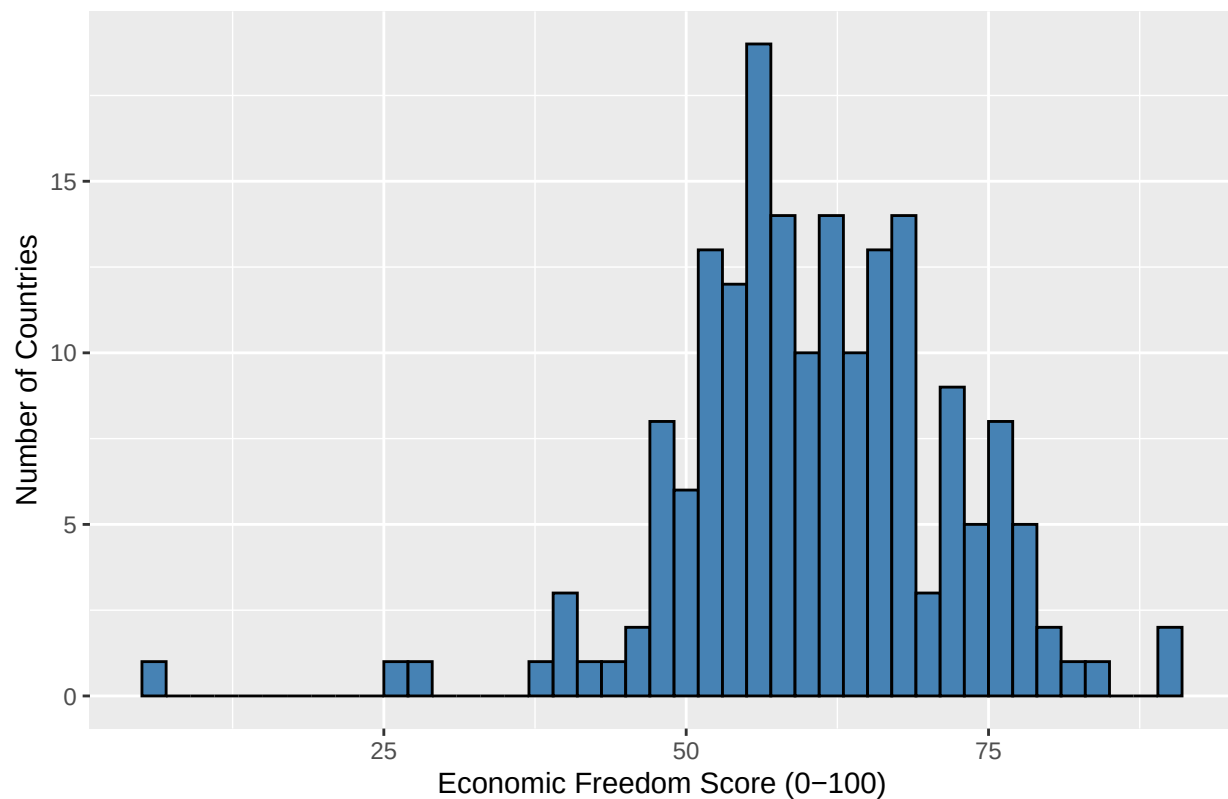
summary(summary_vars)
```

##	X2019.Score	Property.Rights	Judicial.Effectiveness	Government.Integrity
##	Min. : 5.90	Min. : 7.60	Min. : 5.0	Min. : 7.90
##	1st Qu.:53.95	1st Qu.:37.00	1st Qu.:31.0	1st Qu.:27.20
##	Median :60.75	Median :50.10	Median :42.9	Median :35.50
##	Mean :60.77	Mean :52.33	Mean :44.9	Mean :41.47
##	3rd Qu.:67.80	3rd Qu.:65.90	3rd Qu.:54.7	3rd Qu.:50.30
##	Max. :90.20	Max. :97.40	Max. :92.4	Max. :96.70
##	NA's :6	NA's :1	NA's :1	NA's :1
##	Trade.Freedom	GDP.per.Capita..PPP.	GDP.Growth.Rate...	Population..Millions.
##	Min. : 0.00	Min. : 677	Min. : -14.00	Min. : 0.10
##	1st Qu.:66.65	1st Qu.: 4480	1st Qu.: 1.80	1st Qu.: 2.70
##	Median :76.10	Median : 12698	Median : 3.20	Median : 8.80
##	Mean :74.26	Mean : 20757	Mean : 3.47	Mean : 40.16
##	3rd Qu.:84.30	3rd Qu.: 29509	3rd Qu.: 4.65	3rd Qu.: 29.50
##	Max. :95.00	Max. :124529	Max. : 70.80	Max. :1390.10
##	NA's :4	NA's :4	NA's :2	NA's :1

Distribution of Economic Freedom Index Scores

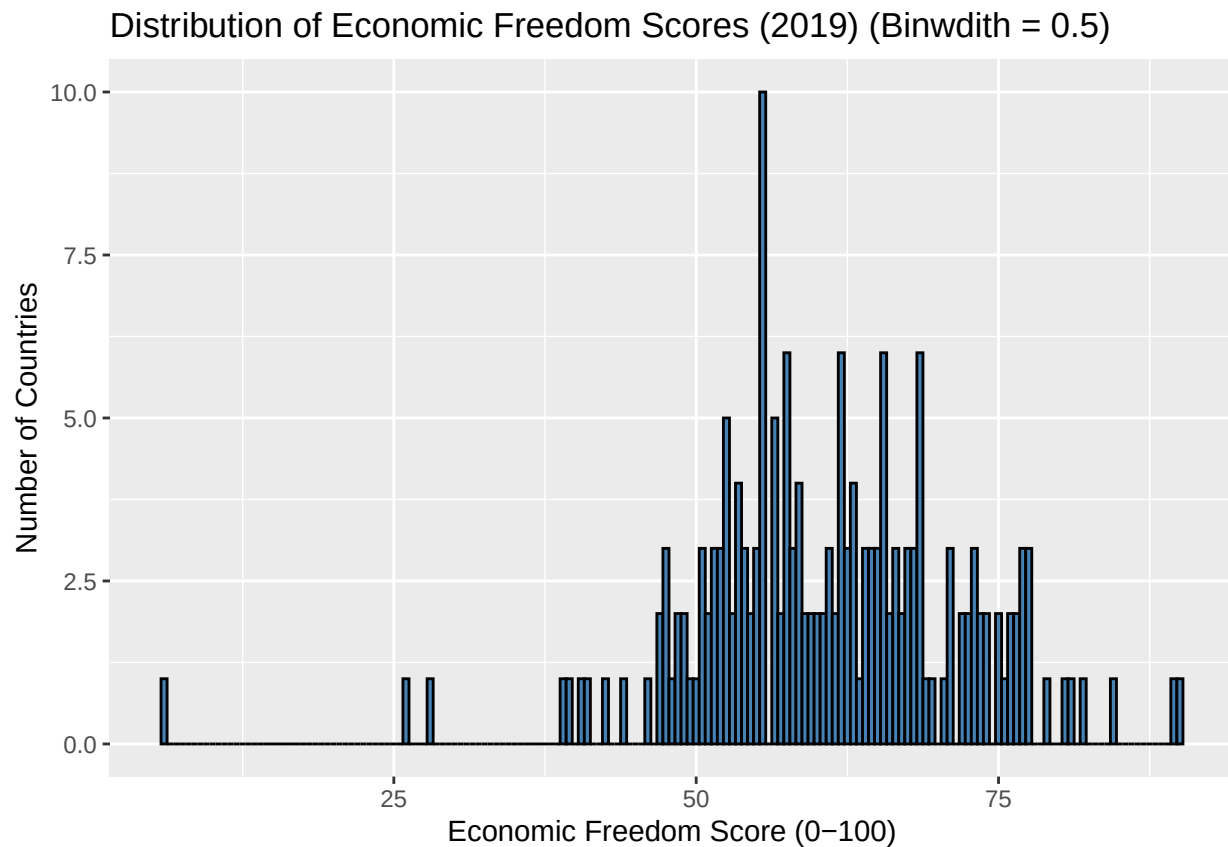
```
ggplot(economic_freedom_data, aes(x = X2019.Score)) +
  geom_histogram(binwidth = 2, fill = "steelblue", color = "black") +
  labs(
    title = "Distribution of Economic Freedom Scores (2019)",
    x = "Economic Freedom Score (0-100)",
    y = "Number of Countries"
  )
```

Distribution of Economic Freedom Scores (2019)



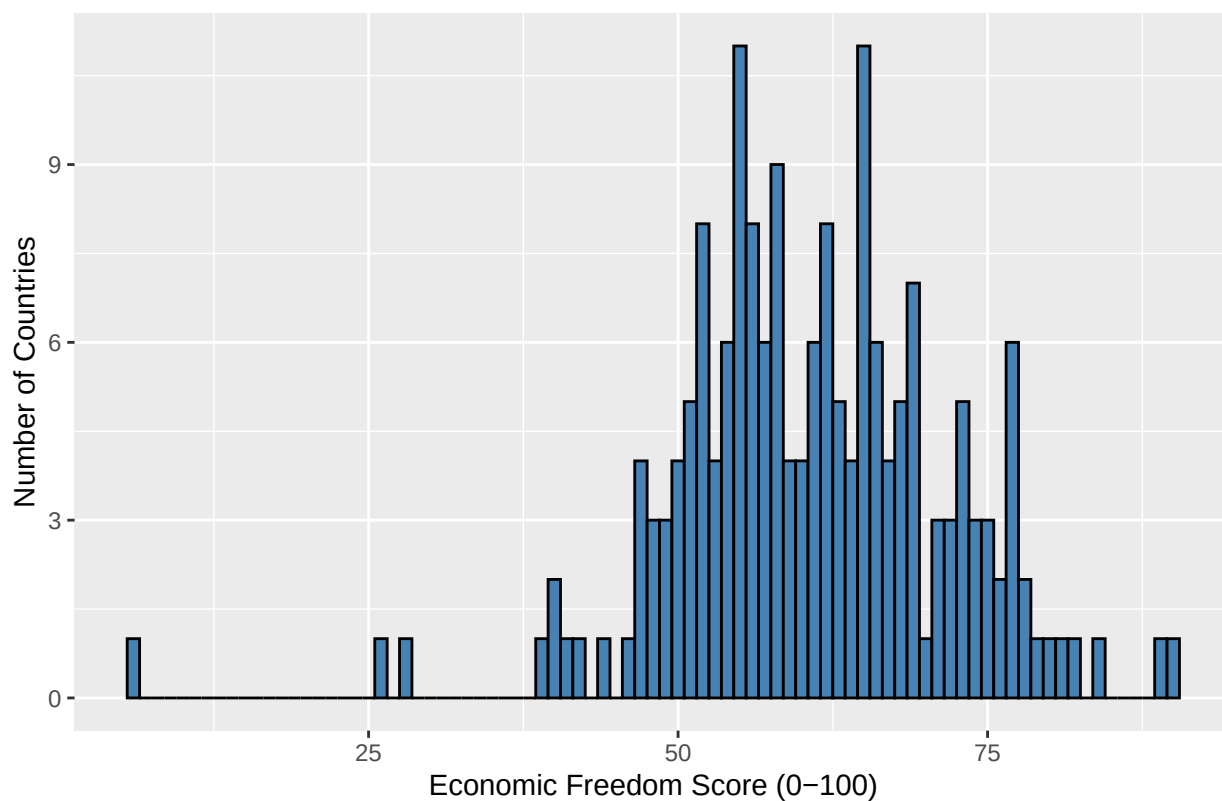
Exploring Different Binwidths for the Histogram of Economic Freedom Scores

```
ggplot(economic_freedom_data, aes(x = X2019.Score)) +  
  geom_histogram(binwidth = 0.5, fill = "steelblue", color = "black") +  
  labs(  
    title = "Distribution of Economic Freedom Scores (2019) (Binwidth = 0.5)",  
    x = "Economic Freedom Score (0-100)",  
    y = "Number of Countries"  
  )
```



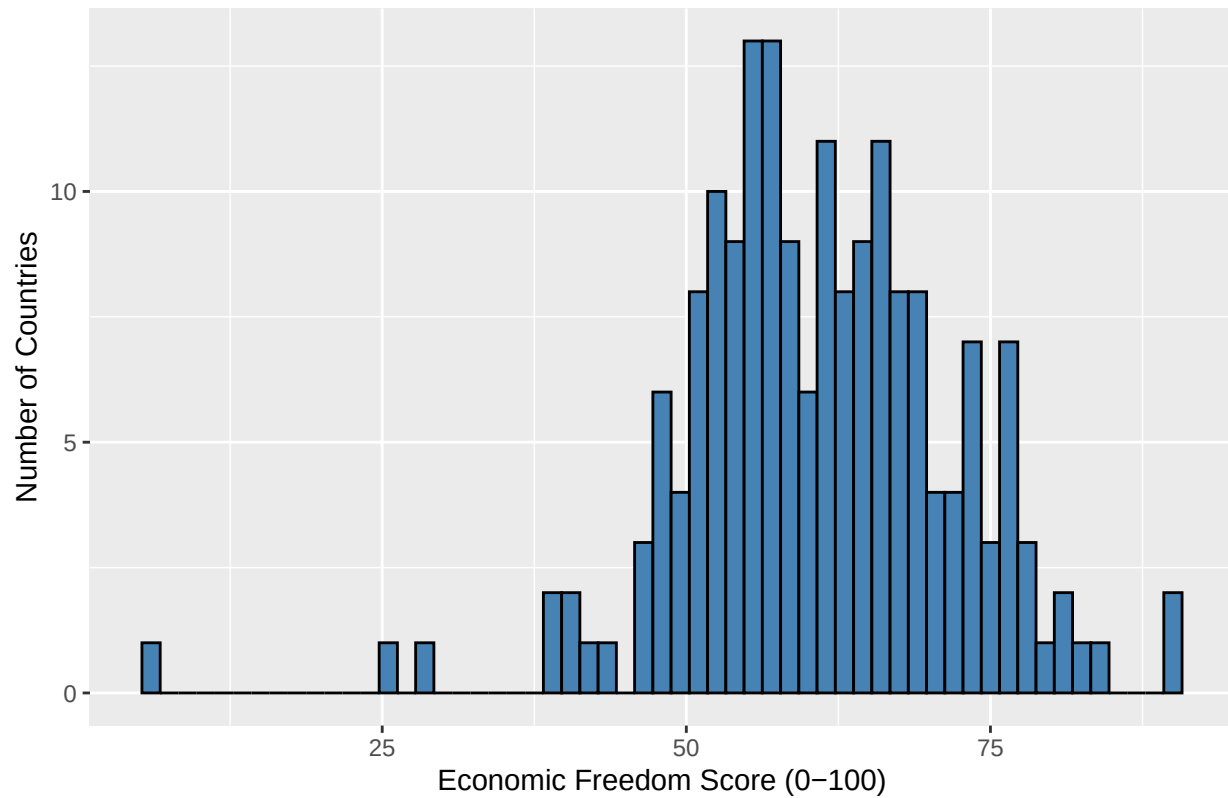
```
ggplot(economic_freedom_data, aes(x = X2019.Score)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "black") +
  labs(
    title = "Distribution of Economic Freedom Scores (2019) (Binwidth = 1)",
    x = "Economic Freedom Score (0-100)",
    y = "Number of Countries"
  )
```

Distribution of Economic Freedom Scores (2019) (Binwidth = 1)



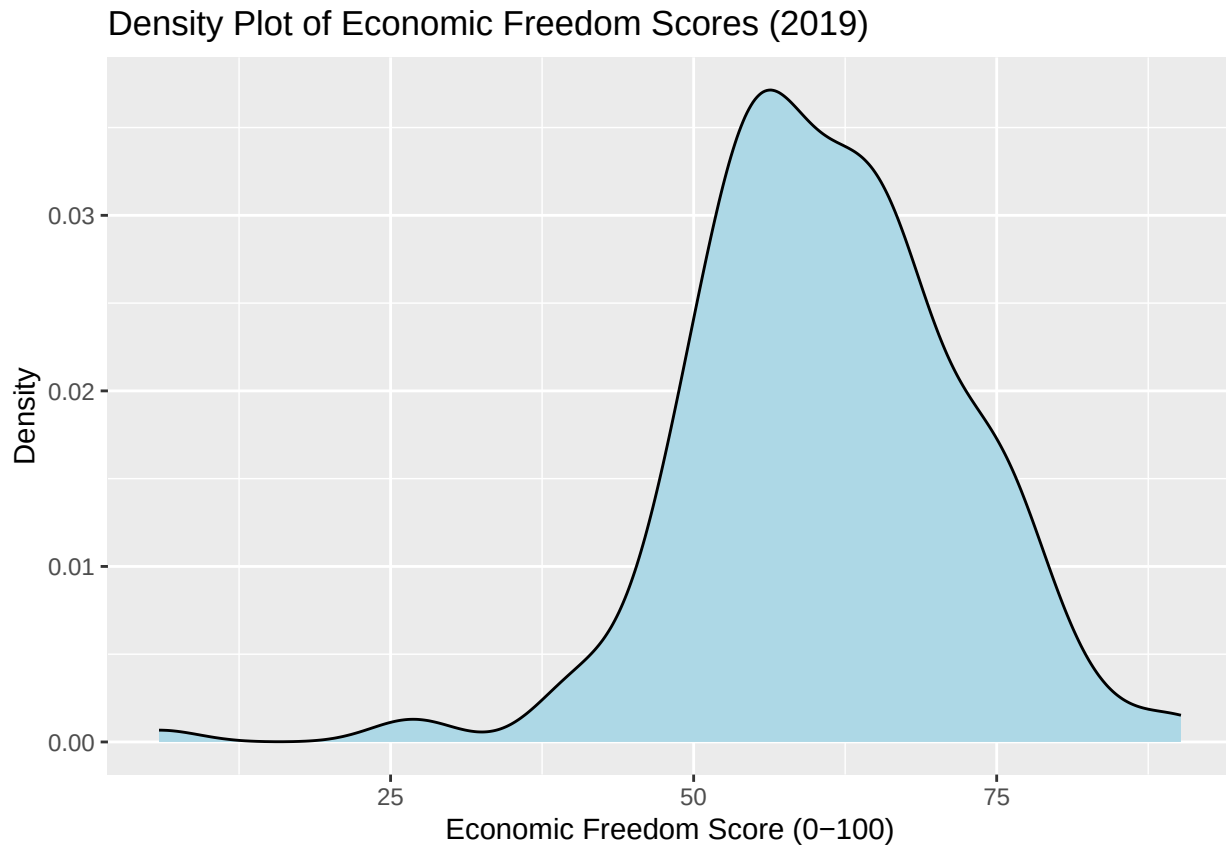
```
ggplot(economic_freedom_data, aes(x = X2019.Score)) +
  geom_histogram(binwidth = 1.5, fill = "steelblue", color = "black") +
  labs(
    title = "Distribution of Economic Freedom Scores (2019) (Binwidth = 1.5)",
    x = "Economic Freedom Score (0-100)",
    y = "Number of Countries"
  )
```

Distribution of Economic Freedom Scores (2019) (Binwidth = 1.5)



Density Plot of Economic Freedom Scores

```
ggplot(economic_freedom_data, aes(x = X2019.Score)) +
  geom_density(fill = "lightblue") +
  labs(
    title = "Density Plot of Economic Freedom Scores (2019)",
    x = "Economic Freedom Score (0-100)",
    y = "Density"
  )
```



Conclusions

Histogram Analysis with Different Binwidths:

Binwidth = 0.5: The histogram shows a more detailed view which reveals small fluctuations in the distribution of economic freedom scores. It would be most useful when exploring fine-grained patterns or checking for small clusters or gaps in the data. Binwidth = 1: The histogram provides a clearer view of the overall shape of the distribution without excessive noise. It would be most useful when exploring descriptive summaries and initial visual analyses. Binwidth = 1.5: The histogram becomes smoother, making overall trends, such as the peak around the global mean and the slight right skew, easier to see. Most countries scores exist from 50 to 75 with less countries scoring outside that range. It would be most useful when presenting general patterns or summarizing findings.

Density Plot Analysis:

The density plot provides a smooth estimate of the distribution of economic freedom scores. It shows that the bulk of countries are centered around a score of approximately 60, with the distribution gradually tapering off toward both higher and lower values. The plot confirms that the distribution is slightly right-skewed, indicating that there are relatively few countries with very high freedom scores (80+), while a large number of countries fall within the moderate range (50-70).

Insights and Implications:

Economic Freedom Concentration: The histograms and density plot reveal that the majority of countries have economic freedom scores clustered between 50 and 70, suggesting that the global economic freedom is generally moderate. Relatively few countries fall at the extremes of the distribution, with only a handful achieving very high scores above 80 or very low scores below 40.

Right-Skewed Distribution: The distribution is slightly right-skewed, indicating that while some nations experience exceptionally high levels of economic freedom, many more remain below those top levels. This

pattern may reflect structural or institutional challenges, such as weak property rights, government intervention, or limited judicial effectiveness. These challenges may prevent countries from reaching the highest tiers of freedom.

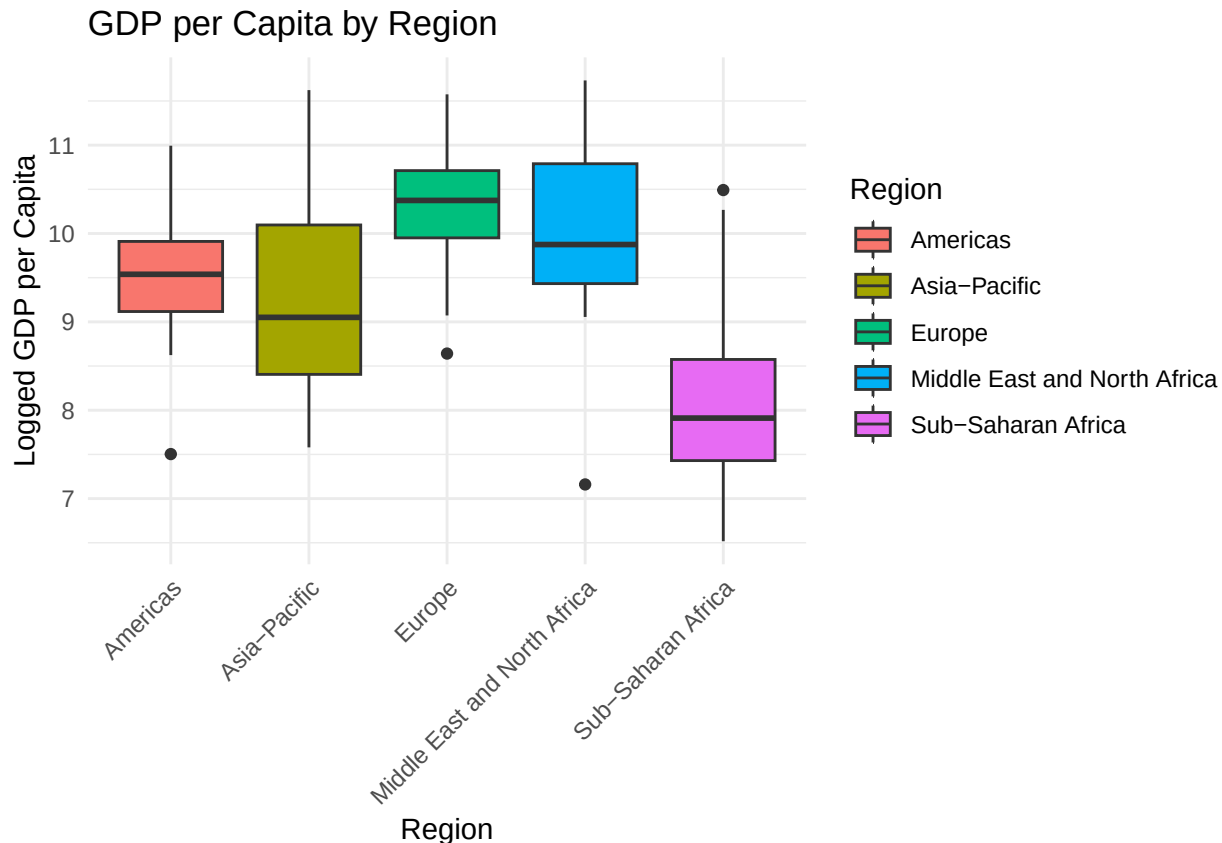
Policy Implications:

Since most countries cluster in the mid-range of economic freedom, policy efforts should prioritize institutional and regulatory reforms that enhance transparency, strengthen judicial independence, and promote open trade. Incremental improvements in these areas could help shift the bulk of nations toward higher freedom scores and, in turn, stimulate greater economic growth and opportunity.

GDP Per Capita by Region

```
# Create logged GDP per capita variable
economic_freedom_data <- economic_freedom_data %>%
  mutate(Logged_GDP_per_Capita = log(GDP.per.Capita..PPP.))

# Boxplot of GDP per Capita by Region
ggplot(economic_freedom_data,
       aes(x = reorder(Region, -Logged_GDP_per_Capita),
           y = Logged_GDP_per_Capita, fill = Region)) +
  geom_boxplot() +
  labs(
    title = "GDP per Capita by Region",
    x = "Region",
    y = "Logged GDP per Capita"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



From the boxplot analysis of GDP per capita by region, we can draw the following conclusions:

Significant Regional Difference: There are clear differences in GDP per capita across regions. Europe and the Middle East and North Africa have the highest median GDP per capita, indicating that wealthier economies are concentrated in these regions. Sub-Saharan Africa, by contrast, has the lowest median, suggesting limited economic development relative to other parts of the world.

High Variability in Some Regions: The Asia-Pacific and Middle East and North Africa regions display the widest spreads, showing that while some countries (e.g., Singapore, Qatar) are very wealthy, others in the same region remain far below the regional averages. This reflects substantial within-region economic inequality.

Regions with Lower GDP per Capita: Sub-Saharan Africa stands out with a consistently lower GDP per capita and a wide range, implying both lower overall wealth and uneven development among its countries. The Americas show mid-range values, with both high-income and developing economic represented.

Outliers: A few outliers appear above the main boxes, representing countries with exceptionally high GDP per capita relative to their regional peers (likely advanced financial or resource-driven economies).

Conclusion: The boxplot highlights pronounced global inequality in income distribution across regions. Wealthier regions such as Europe and the Middle East cluster toward higher GDP per capita, while Sub-Saharan Africa remains substantially below. The broad spread in regions like Asia-Pacific suggests diverse levels of development, combining both emerging and advanced economies. These findings provide important context for interpreting regional differences in economic freedom, as wealth and institutional strength often reinforce each other.

Correlation Analysis

```
cor_matrix <- economic_freedom_data %>%
  select(X2019.Score,
         Property.Rights,
         Judicial.Effectiveness,
         Government.Integrity,
         Trade.Freedom,
         GDP.per.Capita..PPP.,
         GDP.Growth.Rate....,
         Population..Millions.) %>%
  cor(use = "complete.obs")

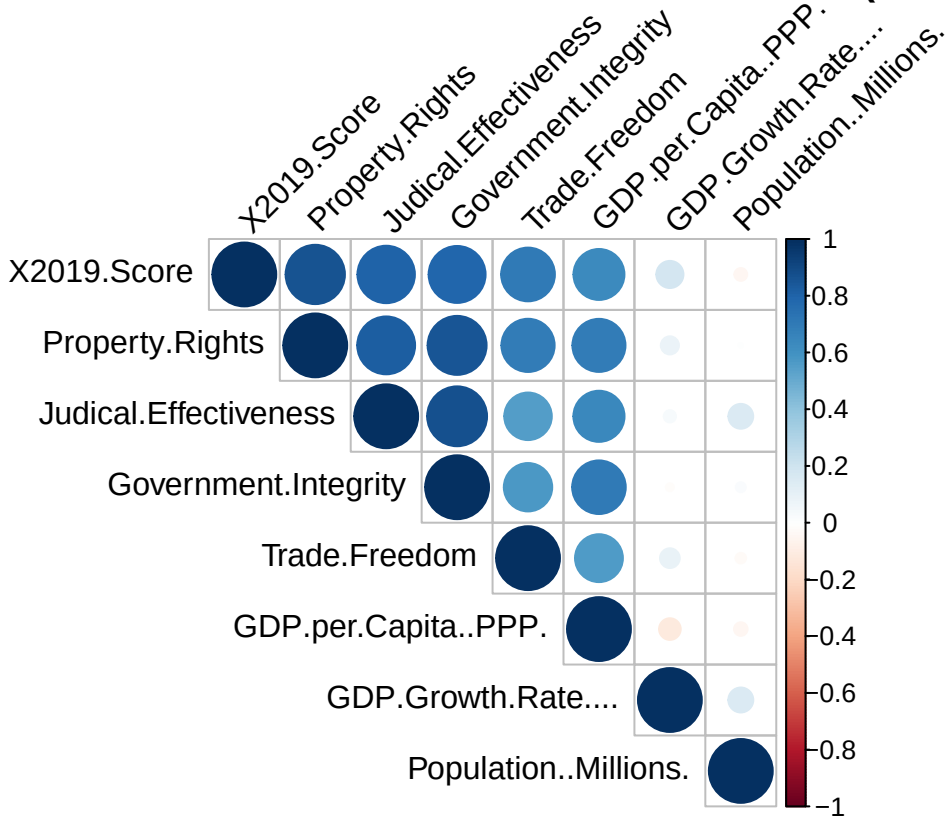
# Round to 2 decimals
round(cor_matrix, 2)
```

```
##           X2019.Score Property.Rights Judicial.Effectiveness
## X2019.Score           1.00           0.86           0.81
## Property.Rights       0.86           1.00           0.82
## Judicial.Effectiveness 0.81           0.82           1.00
## Government.Integrity   0.80           0.85           0.88
## Trade.Freedom          0.71           0.69           0.55
## GDP.per.Capita..PPP.    0.64           0.69           0.64
## GDP.Growth.Rate....    0.18           0.08           0.04
## Population..Millions.  -0.04           0.00           0.15
##
##           Government.Integrity Trade.Freedom GDP.per.Capita..PPP.
## X2019.Score           0.80           0.71           0.64
## Property.Rights       0.85           0.69           0.69
## Judicial.Effectiveness 0.88           0.55           0.64
## Government.Integrity   1.00           0.57           0.70
## Trade.Freedom          0.57           1.00           0.56
## GDP.per.Capita..PPP.    0.70           0.56           1.00
## GDP.Growth.Rate....    -0.02           0.10          -0.11
## Population..Millions.   0.02          -0.03          -0.05
##
##           GDP.Growth.Rate.... Population..Millions.
## X2019.Score           0.18          -0.04
## Property.Rights       0.08           0.00
## Judicial.Effectiveness 0.04           0.15
## Government.Integrity  -0.02           0.02
## Trade.Freedom          0.10          -0.03
## GDP.per.Capita..PPP.   -0.11          -0.05
## GDP.Growth.Rate....    1.00           0.15
## Population..Millions.   0.15           1.00
```

Correlation Analysis with Corrplot

```
corrplot(cor_matrix,
         method = "circle",
         type = "upper",
         tl.col = "black",
         tl.srt = 45,
         title = "Correlation Matrix of Economic Freedom Factors (2019)",
         mar = c(0, 0, 1, 0))
```

Correlation Matrix of Economic Freedom Factors (2019)



Correlation Analysis and Conclusions

Interpretation of the Correlation Analysis:

Property Rights and Economic Freedom:

There is a strong positive correlation ($r = 0.86$) between Property Rights and Economic Freedom. This suggests that nations with well-established legal protections for private ownership, contract enforcement, and secure property systems tend to experience higher levels of overall economic freedom. The protection of property rights appears to be one of the strongest institutional pillars supporting free markets.

Judicial Effectiveness and Economic Freedom:

A high positive correlation ($r = 0.81$) indicates that countries with efficient and impartial judicial systems tend to have greater economic freedom. Judicial effectiveness reinforces the rule of law by ensuring contracts are honored and corruption is limited, fostering investor confidence and fair competition.

Government Integrity and Economic Freedom:

The correlation between Government Integrity and Economic Freedom is strong ($r = 0.80$). This demonstrates that transparency and low corruption levels are closely tied to freer economic systems. When citizens and businesses can trust public institutions, economic activity is less distorted by rent-seeking and political favoritism.

Trade Freedom and Economic Freedom:

The relationship between Trade Freedom and overall Economic Freedom is moderately strong ($r = 0.71$). Countries that maintain open trade policies, such as lower tariffs, minimal quotas, and fewer regulatory barriers, tend to perform better in terms of overall economic freedom. This reinforces the idea that integration into global markets supports freer domestic economies.

GDP per Capita (PPP) and Economic Freedom:

There is a moderately positive correlation ($r = 0.64$) between GDP per capita and Economic Freedom. Wealthier nations generally enjoy higher economic freedom, reflecting how sustained prosperity and robust institutions reinforce one another. However, the relationship is not perfect. Some resource-rich or reforming nations may have higher GDP levels without matching levels of institutional freedom.

GDP Growth Rate and Economic Freedom:

A weak positive correlation ($r = 0.18$) suggests that short-term economic growth does not strongly predict higher economic freedom. While freer economies may grow more steadily in the long term, annual growth rates can fluctuate due to external shocks, reforms, or commodity price cycles.

Population and Economic Freedom

There is virtually no correlation ($r = -0.04$) between population size and economic freedom. This implies that demographic size alone does not determine how free an economy is. Both small and large nations can achieve either strong or weak institutional performance depending on their governance structures.

Conclusions from the Correlation Analysis

The results indicate that institutional quality variables, particularly Property Rights, Judicial Effectiveness, and Government Integrity, are the strongest correlates of economic freedom. Trade Freedom and GDP per Capital also show meaningful positive relationships, suggesting that open markets and wealth accumulation reinforce one another. By contrast, GDP Growth Rate and Population exhibit weak or negligible correlations, highlighting that economic freedom depends more on long-term institutional foundations than on short-term economic performance or demographic factors.

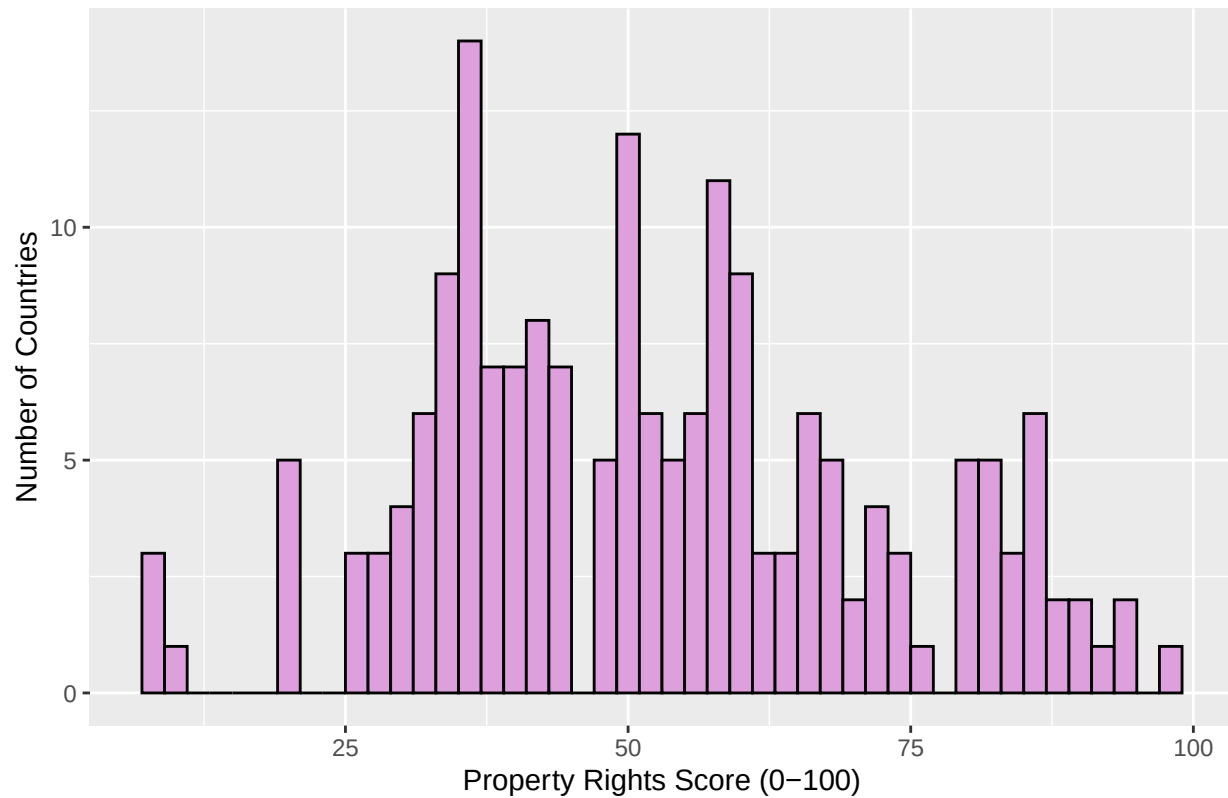
Policy Implications

These results highlight the importance of strengthening institutions to enhance economic freedom. Policies that protect property rights, improve judicial effectiveness, reduce corruption, and promote open trade are most effective in fostering long-term prosperity. Ultimately, strong and transparent governance services as the foundation for sustainable economic growth and greater overall freedom.

Distribution of Property Rights Scores

```
ggplot(economic_freedom_data, aes(x = Property.Rights)) +  
  geom_histogram(binwidth = 2, fill = "plum", color = "black") +  
  labs(  
    title = "Distribution of Property Rights Scores",  
    x = "Property Rights Score (0-100)",  
    y = "Number of Countries"  
  )
```

Distribution of Property Rights Scores



Exploring Different Binwidths for the Histogram of Property Rights Scores

```
ggplot(economic_freedom_data, aes(x = Property.Rights)) +  
  geom_histogram(binwidth = 1, fill = "plum", color = "black") +  
  labs(  
    title = "Distribution of Property Rights Scores (binwidth = 1)",  
    x = "Property Rights Score (0-100)",  
    y = "Number of Countries"  
  )
```

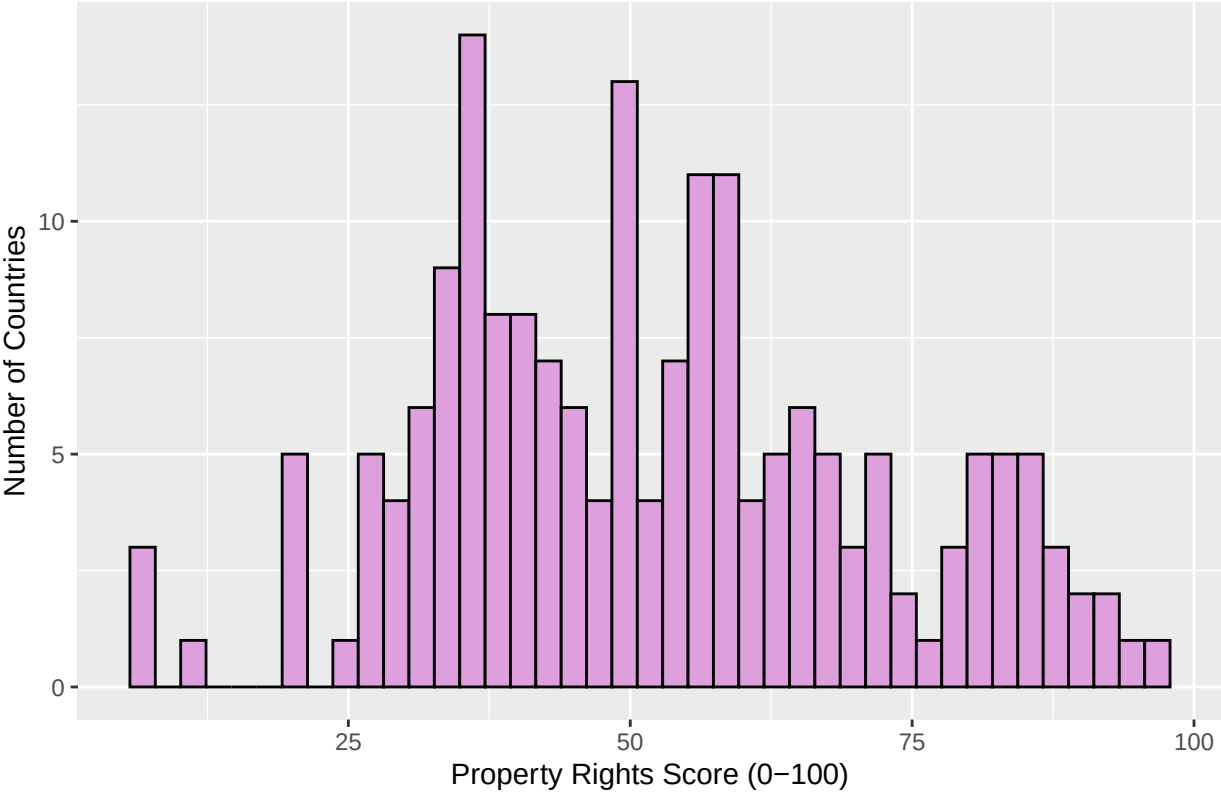


```
ggplot(economic_freedom_data, aes(x = Property.Rights)) +  
  geom_histogram(binwidth = 1.75, fill = "plum", color = "black") +  
  labs(  
    title = "Distribution of Property Rights Scores (binwidth = 1.75)",  
    x = "Property Rights Score (0-100)",  
    y = "Number of Countries"  
  )
```



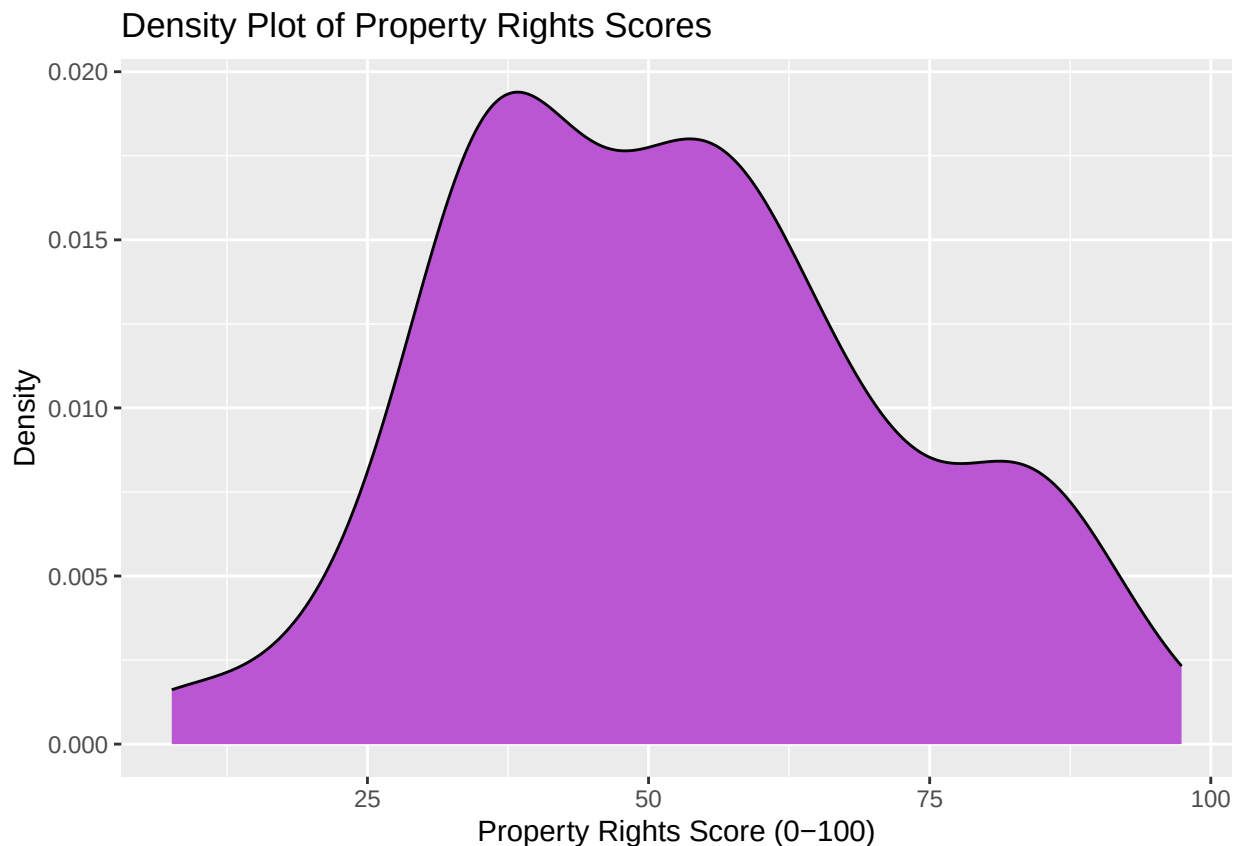
```
ggplot(economic_freedom_data, aes(x = Property.Rights)) +  
  geom_histogram(binwidth = 2.25, fill = "plum", color = "black") +  
  labs(  
    title = "Distribution of Property Rights Scores (binwidth = 2.25)",  
    x = "Property Rights Score (0-100)",  
    y = "Number of Countries"  
  )
```

Distribution of Property Rights Scores (binwidth = 2.25)



Density Plot of Property Rights Scores

```
ggplot(economic_freedom_data, aes(x = Property.Rights)) +  
  geom_density(fill = "mediumorchid") +  
  labs(  
    title = "Density Plot of Property Rights Scores",  
    x = "Property Rights Score (0-100)",  
    y = "Density"  
  )
```



Conclusions

Histogram Analysis with Different Binwidths:

Binwidth = 1: The histogram shows a more detailed view of the Property Rights distributions, showing small fluctuations and minor peaks across the score range. However, the high granularity makes it harder to see the overall shape of the distribution. Binwidth = 1.75: The histogram strikes a balance between precision and clarity. It highlights that most countries cluster between 40 and 65 on the Property Rights scale, with fewer countries at either extreme. Binwidth = 2.25: The histogram smooths out variation and reveals the main pattern that Property Rights scores are concentrated around mid-range values. Only a small number of countries achieve a very high (above 80) or very low (below 25) score.

Density Plot Analysis:

The density plot provides a smooth representation of the Property Rights distribution. It shows a clear concentration of countries around the mid-range (40-60), tapering off toward both ends. The slight double-peaked shape suggests a possible bimodal pattern, indicating that while some nations have moderately strong property protections, others remain considerably weaker in safeguarding ownership and enforcement rights.

Insights and Implications:

Property Rights Concentration: The histograms and density plot reveal that the majority of countries maintain moderate property rights with scores clustered between 40 and 60. This suggests that while property ownership is broadly recognized, consistent enforcement and legal protection still lag globally.

Outliers: A smaller number of nations exhibit very low or high scores, reflecting stark institutional contrasts. Countries range from weak legal systems and corruption issues to highly developed judicial protections.

Policy Implications: Strengthening property rights should be a key institutional priority especially in developing economies. Policies aimed at improving judicial enforcement, transparency, and corruption control can help nations move toward stronger, more reliable property systems that foster improvement and economic freedom.

Explanation of the Two Modes:

The presence of two modes (or peaks) in the density plot of property right scores suggests that the distribution of property right scores is bimodal. This indicates that there are two distinct groups of countries with different typical property right score levels. Here's how to interpret this:

First Mode (Lower Property Right Scores): The first peak, typically around scores of 30-45, represents countries with weaker property right institutions. These nations often struggle with inconsistent legal enforcement, high levels of corruption, or unclear land ownership systems. In such environment, citizens and businesses may face difficulties securing ownership titles or resolving disputes, discouraging both domestic and foreign investment. Contributing factors may include: weak judicial systems and limited rule of law, government interference or expropriation risks, inefficient land registries or bureaucratic delays.

Second Mode (Higher Property Right Scores): The second peak, typically appearing between 65-80, represents countries with well-established and consistently enforce property rights. These nations tend to have transparent legal systems, efficient courts, and stable governments that uphold ownership and contractual agreements. Common characteristics include: strong judicial independence and anti-corruption frameworks, secure land tenure and low expropriation risk, institutional reliability that supports long-term economic growth.

Research Questions

What are the top 5 countries with the highest levels of economic freedom in 2019?

```
# Selecting the top 5 countries
```

```
top5 <- economic_freedom_data %>%  
  arrange(desc(X2019.Score)) %>%  
  select(Country.Name, X2019.Score) %>%  
  head(5)
```

```
top5
```

```
##   Country.Name X2019.Score  
## 1    Hong Kong      90.2  
## 2    Singapore      89.4  
## 3   New Zealand      84.4  
## 4   Switzerland      81.9  
## 5    Australia      80.9
```

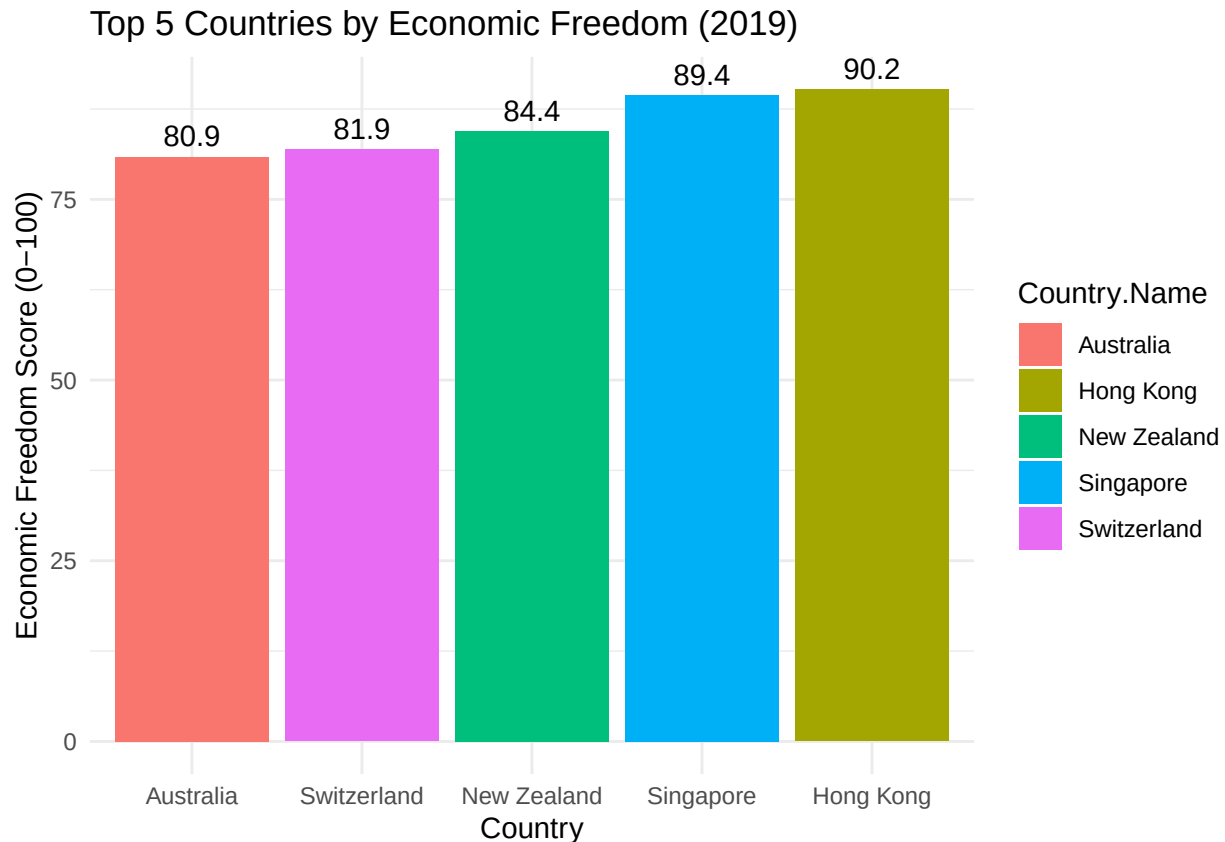
```
# Visualizing the top 5 countries
```

```
ggplot(top5, aes(x = reorder(Country.Name, X2019.Score), y = X2019.Score, fill = Country.Name)) +  
  geom_bar(stat = "identity") +  
  geom_text(aes(label = X2019.Score), vjust = -0.5, color = "black", size = 4) +  
  theme_minimal() +  
  labs()
```

```

title = "Top 5 Countries by Economic Freedom (2019)",
x = "Country",
y = "Economic Freedom Score (0-100)"
)

```



What are the top 5 countries with the lowest levels of economic freedom in 2019?

```
# Selecting the bottom 5 countries
```

```

bottom5 <- economic_freedom_data %>%
  arrange(X2019.Score) %>%
  select(Country.Name, X2019.Score) %>%
  head(5)

```

```
bottom5
```

```

##      Country.Name X2019.Score
## 1      Korea, North      5.9
## 2      Venezuela      25.9
## 3         Cuba      27.8
## 4       Eritrea      38.9
## 5 Congo, Republic of    39.7

```

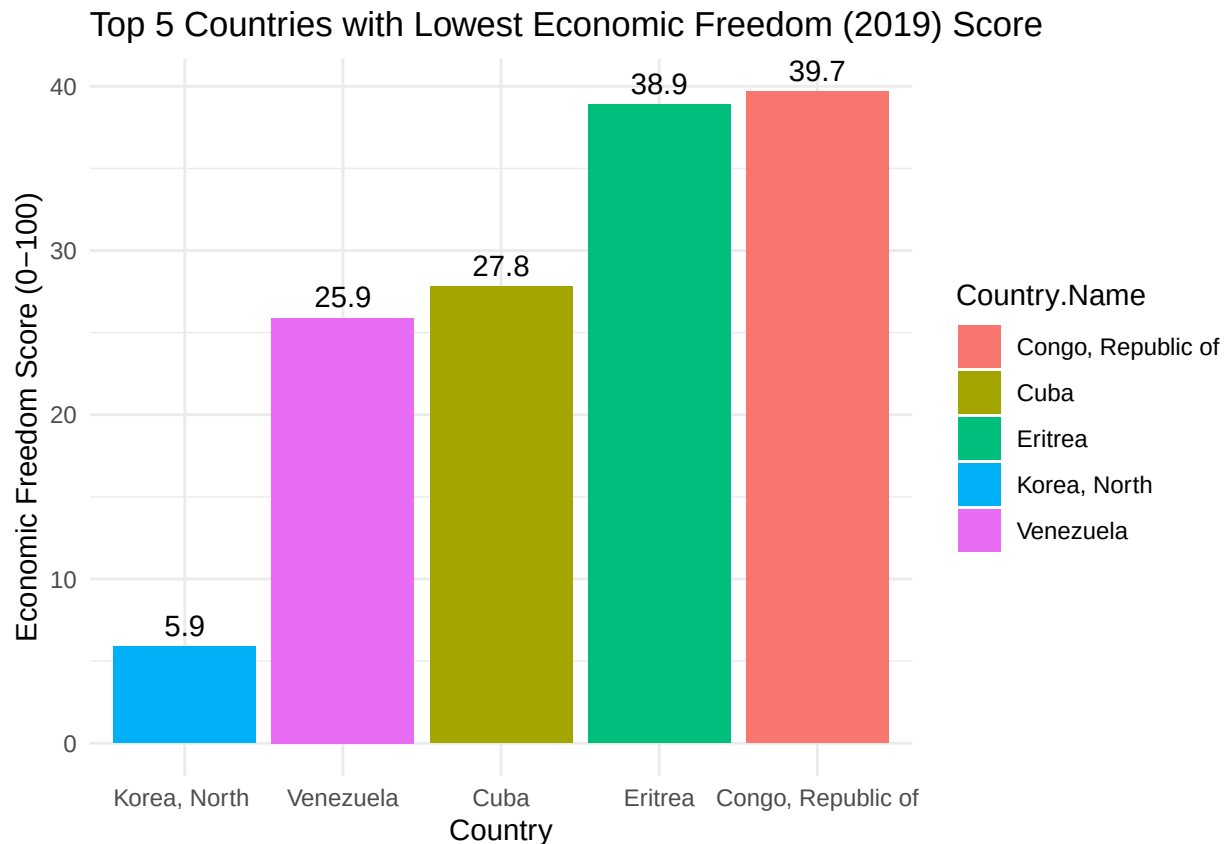
```
# Visualizing the bottom 5 countries
```

```

ggplot(bottom5, aes(x = reorder(Country.Name, X2019.Score), y = X2019.Score, fill = Country.Name)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = X2019.Score), vjust = -0.5, color = "black", size = 4) +

```

```
theme_minimal() +
labs(
  title = "Top 5 Countries with Lowest Economic Freedom (2019) Score",
  x = "Country",
  y = "Economic Freedom Score (0-100)"
)
```



How does GDP per capita relate to economic freedom?

Finding correlation between GDP per capita and economic freedom

```
gdp_economic_freedom_cor = cor(economic_freedom_data$X2019.Score,
  economic_freedom_data$GDP.per.Capita..PPP.,
  use = "complete.obs")
```

```
gdp_economic_freedom_cor
```

```
## [1] 0.6365604
```

Plotting scatterplot of GDP per capita vs economic freedom

```
gdp_freedom_plot <- ggplot(economic_freedom_data,
  aes(x = GDP.per.Capita..PPP., y = X2019.Score)) +
  geom_point(aes(color = Region), size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "GDP per Capita (PPP) vs Economic Freedom Score (2019)",
    x = "GDP per Capita (PPP, in $)",
```

```

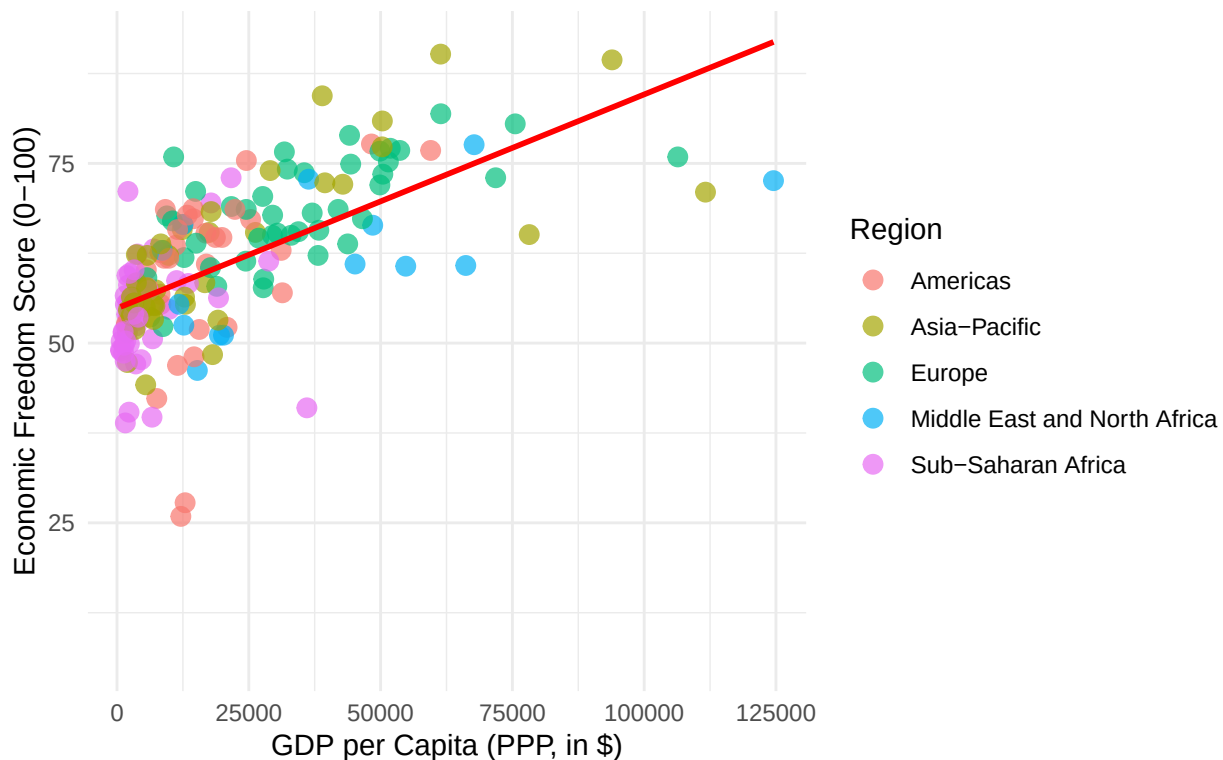
y = "Economic Freedom Score (0-100)",
caption = "Source: Heritage Foundation, Economic Freedom Index 2019"
) +
theme_minimal()

gdp_freedom_plot

```

```
## `geom_smooth()` using formula = 'y ~ x'
```

GDP per Capita (PPP) vs Economic Freedom Score (2019)



Source: Heritage Foundation, Economic Freedom Index 2019

Statistical Summary of the Relationship

We can further analyze the relationship by dividing the countries into groups based on their GDP per capita. This will allow us to compare happiness scores across wealth tiers.

```

# Create GDP per capita categories (quartiles)
economic_freedom_data <- economic_freedom_data %>%
  mutate(GDP_Category = cut(GDP.per.Capita..PPP.,
                           breaks = quantile(GDP.per.Capita..PPP.,
                                             probs = seq(0, 1, 0.25),
                                             na.rm = TRUE),
                           include.lowest = TRUE,
                           labels = c("Low", "Lower-Middle",
                                      "Upper-Middle", "High")))

# Summary of Economic Freedom by GDP category
gdp_freedom_summary <- economic_freedom_data %>%
  group_by(GDP_Category) %>%
  summarise(

```

```

Mean_Freedom = mean(X2019.Score, na.rm = TRUE),
SD_Freedom = sd(X2019.Score, na.rm = TRUE),
Min_Freedom = min(X2019.Score, na.rm = TRUE),
Max_Freedom = max(X2019.Score, na.rm = TRUE)
)

# Display the summary table
gdp_freedom_summary

## # A tibble: 5 x 5
##   GDP_Category Mean_Freedom SD_Freedom Min_Freedom Max_Freedom
##   <fct>          <dbl>      <dbl>      <dbl>      <dbl>
## 1 Low           53.7        5.50       38.9       71.1
## 2 Lower-Middle  57.6        8.81       25.9       75.9
## 3 Upper-Middle  61.2        9.02       27.8       75.4
## 4 High          71.5        8.76       41         90.2
## 5 <NA>          5.9         NA         5.9        5.9

```

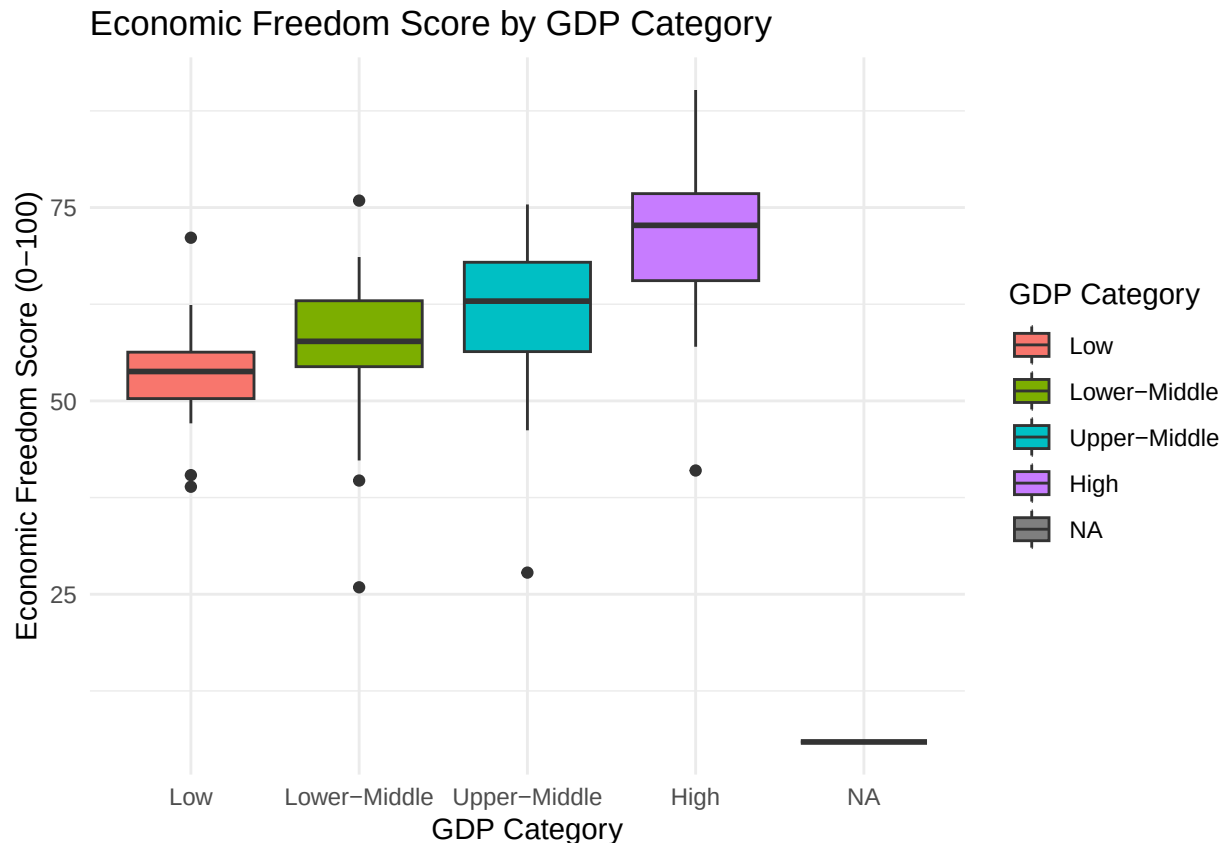
Visualization by GDP Category

To further understand how GDP per capita influences happiness, let's visualize the happiness scores across different GDP categories.

```

ggplot(economic_freedom_data,
       aes(x = GDP_Category, y = X2019.Score, fill = GDP_Category)) +
  geom_boxplot() +
  labs(
    title = "Economic Freedom Score by GDP Category",
    x = "GDP Category",
    y = "Economic Freedom Score (0-100)",
    fill = "GDP Category"
  ) +
  theme_minimal()

```



Conclusion

The results of the analysis show a moderately strong positive correlation ($r = 0.637$) between GDP per capita and economic freedom. This indicates that countries with higher income levels generally tend to exhibit greater economic freedom. The scatterplot clearly illustrates this upward trend, although some variability remains—suggesting that while economic prosperity and freedom often go hand in hand, GDP alone does not fully determine a nation's level of economic freedom.

The summary statistics and boxplot further reinforce this relationship: nations in the High GDP category have the highest average economic freedom scores, while those in the Low GDP group score noticeably lower. However, the presence of variation within each GDP tier implies that additional institutional and policy factors, such as governance integrity, trade openness, and property rights, also play key roles in shaping economic freedom.

Do countries with stronger property rights tend to exhibit higher economic freedom?

```
# Calculate the correlation between Property Rights and Economic Freedom
property_rights_freedom_correlation = cor(economic_freedom_data$Property.Rights,
                                          economic_freedom_data$X2019.Score,
                                          use = "complete.obs")
```

```
# Display the correlation value
property_rights_freedom_correlation
```

```
## [1] 0.8304095
```

```
# Scatter plot: Property Rights vs Economic Freedom
ggplot(economic_freedom_data, aes(x = Property.Rights, y = X2019.Score)) +
  geom_point(color = "mediumorchid", size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
```

```
labs(
  title = "Property Rights vs Economic Freedom Score (2019)",
  x = "Property Rights Score (0-100)",
  y = "Economic Freedom Score (0-100)",
  caption = "Source: Heritage Foundation, Economic Freedom Index 2019"
) +
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Property Rights vs Economic Freedom Score (2019)



Source: Heritage Foundation, Economic Freedom Index 2019

```
# Create categories based on Property Rights scores (split into quintiles)
economic_freedom_data = economic_freedom_data %>%
  mutate(Property_Rights_Category = cut(Property.Rights,
    breaks = quantile(Property.Rights, probs = seq(0, 1, 0.2), na.rm = TRUE),
    include.lowest = TRUE,
    labels = c("Low Property Rights",
               "Lower-Middle Property Rights",
               "Upper-Middle Property Rights",
               "High Property Rights",
               "Very High Property Rights")))

# Summary of economic freedom scores by property rights category
property_rights_freedom_summary = economic_freedom_data %>%
  group_by(Property_Rights_Category) %>%
  summarise(
    Mean_Freedom = mean(X2019.Score, na.rm = TRUE),
    SD_Freedom = sd(X2019.Score, na.rm = TRUE),
```

```

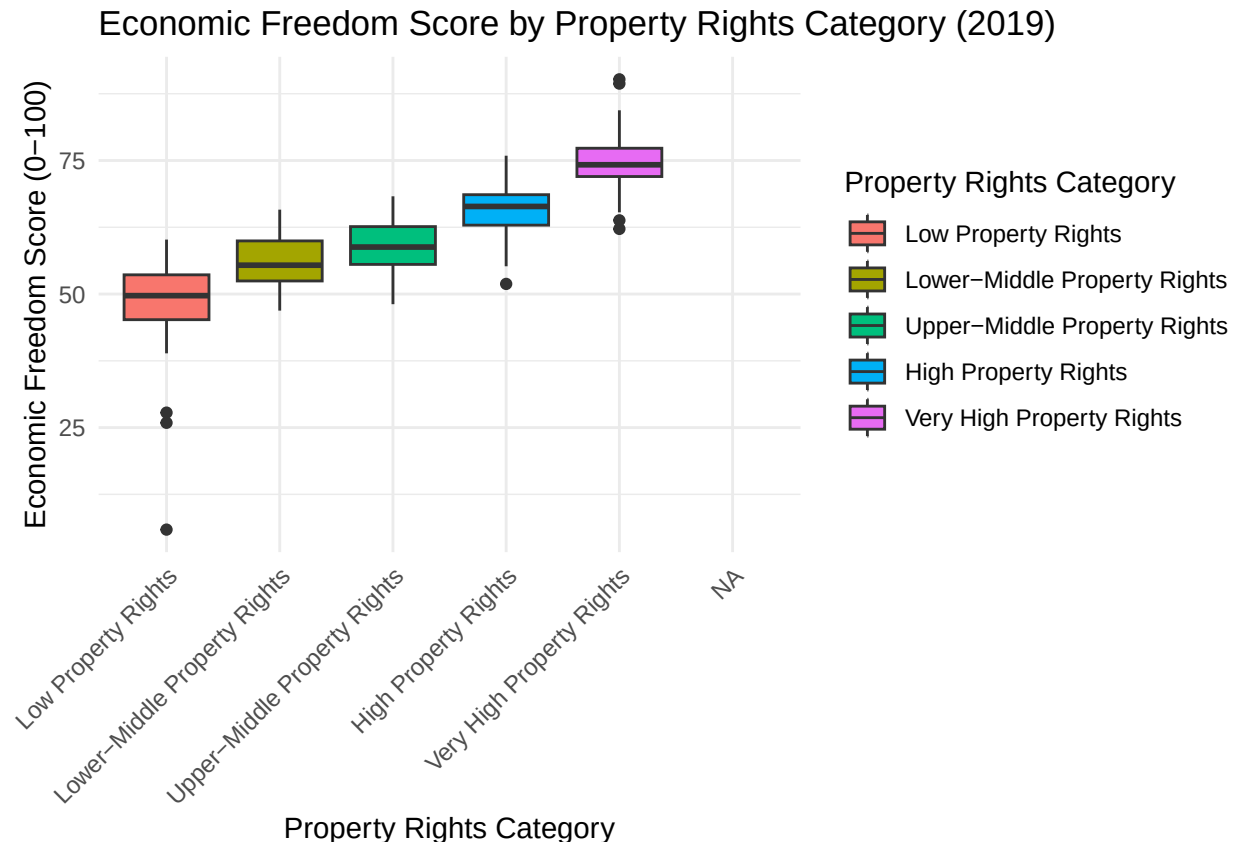
Min_Freedom = min(X2019.Score, na.rm = TRUE),
Max_Freedom = max(X2019.Score, na.rm = TRUE)
)

# Display the summary
property_rights_freedom_summary

## # A tibble: 6 x 5
##   Property_Rights_Category    Mean_Freedom SD_Freedom Min_Freedom Max_Freedom
##   <fct>                <dbl>      <dbl>      <dbl>      <dbl>
## 1 Low Property Rights      47.4        10.5         5.9        60.2
## 2 Lower-Middle Property Rights 56.1         4.72        46.9        65.8
## 3 Upper-Middle Property Rights 59.2         4.68        48.1        68.3
## 4 High Property Rights     65.6         5.26        51.9        75.9
## 5 Very High Property Rights  74.5         6.34        62.2        90.2
## 6 <NA>                   NaN          NA          Inf        -Inf

# Boxplot of Economic Freedom by Property Rights Category
ggplot(economic_freedom_data, aes(x = Property_Rights_Category, y = X2019.Score, fill = Property_Rights_Category)) +
  geom_boxplot() +
  labs(
    title = "Economic Freedom Score by Property Rights Category (2019)",
    x = "Property Rights Category",
    y = "Economic Freedom Score (0-100)",
    fill = "Property Rights Category"
  ) +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



What is the relationship between government integrity and economic freedom?

```
# Calculate the correlation between Government Integrity and Economic Freedom
gov_integrity_freedom_correlation = cor(economic_freedom_data$Government.Integrity,
                                         economic_freedom_data$X2019.Score,
                                         use = "complete.obs")
```

```
# Display the correlation value
gov_integrity_freedom_correlation
```

```
## [1] 0.7637149
```

```
# Scatter plot: Government Integrity vs Economic Freedom
ggplot(economic_freedom_data, aes(x = Government.Integrity, y = X2019.Score)) +
  geom_point(color = "darkgreen", size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(
    title = "Government Integrity vs Economic Freedom Score (2019)",
    x = "Government Integrity Score (0-100)",
    y = "Economic Freedom Score (0-100)",
    caption = "Source: Heritage Foundation, Economic Freedom Index 2019"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Government Integrity vs Economic Freedom Score (2019)



```
# Create categories based on Government Integrity (quintiles)
economic_freedom_data <- economic_freedom_data %>%
  mutate(GovIntegrity_Category = cut(Government.Integrity,
```

```

breaks = quantile(Government.Integrity, probs = seq(0, 1, 0.2), na
include.lowest = TRUE,
labels = c("Low Government Integrity",
           "Lower-Middle Government Integrity",
           "Upper-Middle Government Integrity",
           "High Government Integrity",
           "Very High Government Integrity"))

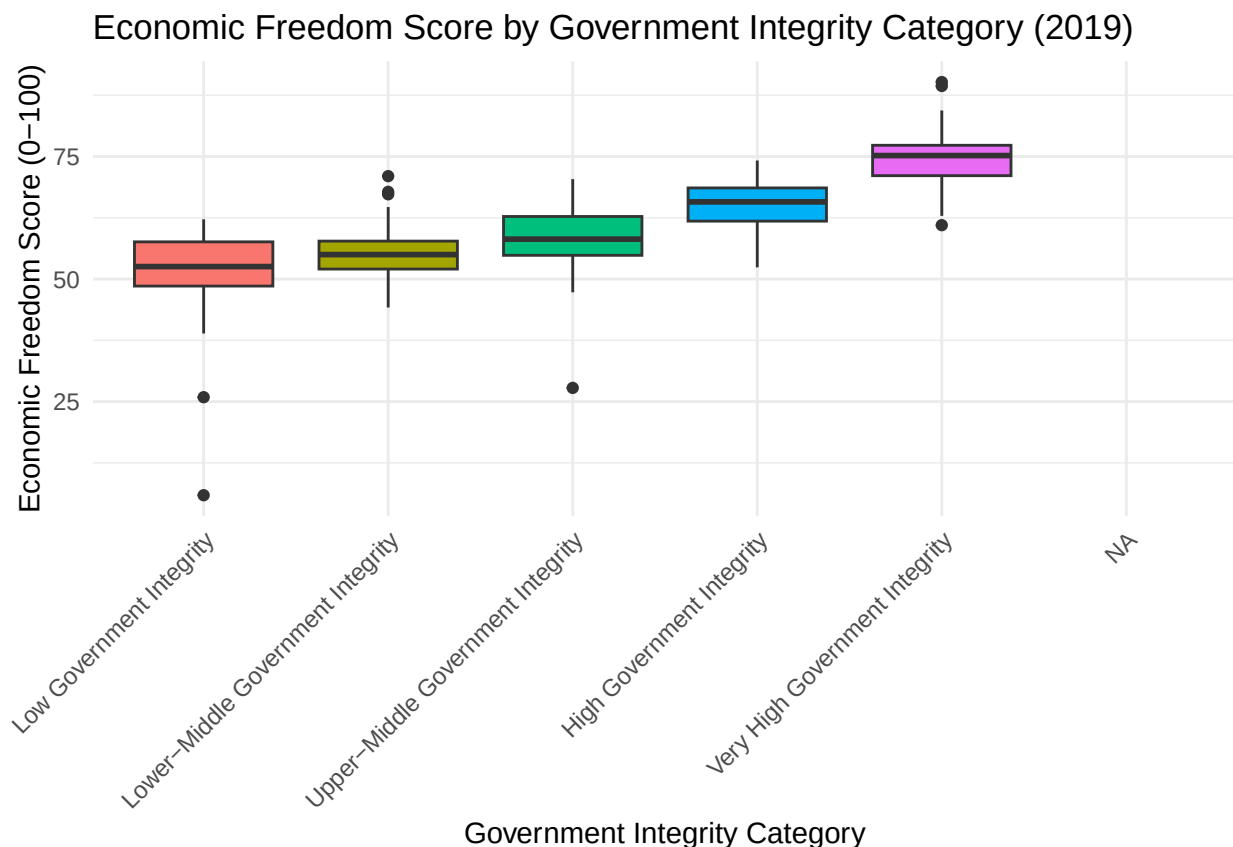
# Summary of Economic Freedom by Government Integrity category
gov_integrity_freedom_summary <- economic_freedom_data %>%
  group_by(GovIntegrity_Category) %>%
  summarise(
    Mean_Freedom = mean(X2019.Score, na.rm = TRUE),
    SD_Freedom   = sd(X2019.Score, na.rm = TRUE),
    Min_Freedom  = min(X2019.Score, na.rm = TRUE),
    Max_Freedom  = max(X2019.Score, na.rm = TRUE)
  )

# Display the summary
gov_integrity_freedom_summary

## # A tibble: 6 x 5
##   GovIntegrity_Category      Mean_Freedom SD_Freedom Min_Freedom Max_Freedom
##   <fct>                  <dbl>      <dbl>      <dbl>      <dbl>
## 1 Low Government Integrity    50.5        11.3         5.9        62.2
## 2 Lower-Middle Government Integ~ 55.4         6.24        44.2         71
## 3 Upper-Middle Government Integ~ 58.2         7.99        27.8        70.4
## 4 High Government Integrity    65.0         5.36        52.4        74.2
## 5 Very High Government Integrity 74.2         6.89         61         90.2
## 6 <NA>                     NaN          NA          Inf         -Inf

# Boxplot of Economic Freedom by Government Integrity category
ggplot(economic_freedom_data,
       aes(x = GovIntegrity_Category, y = X2019.Score, fill = GovIntegrity_Category)) +
  geom_boxplot() +
  labs(
    title = "Economic Freedom Score by Government Integrity Category (2019)",
    x = "Government Integrity Category",
    y = "Economic Freedom Score (0-100)",
    fill = "Government Integrity Category"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")

```



Conclusion

The results of the analysis show a strong positive correlation between government integrity and economic freedom. The scatterplot indicates that countries with higher levels of government integrity tend to have higher economic freedom scores, reflecting the role of transparency and accountability in promoting open markets. The statistical summary and boxplot further confirm this relationship, showing that nations with higher integrity categories consistently achieve higher average economic freedom scores.

Do countries with greater trade openness experience higher economic freedom?

```
# Calculate the correlation between Trade Freedom and Economic Freedom
trade_freedom_correlation <- cor(economic_freedom_data$Trade.Freedom,
                                economic_freedom_data$X2019.Score,
                                use = "complete.obs")

# Display the correlation value
trade_freedom_correlation

## [1] 0.753086

# Scatter plot of Trade Freedom vs Economic Freedom
ggplot(economic_freedom_data, aes(x = Trade.Freedom, y = X2019.Score)) +
  geom_point(color = "brown", size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(
    title = "Trade Freedom vs Economic Freedom Score (2019)",
    x = "Trade Freedom Score (0-100)",
    y = "Economic Freedom Score (0-100)",
    caption = "Source: Heritage Foundation, Economic Freedom Index 2019"
```

```
) +  
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Trade Freedom vs Economic Freedom Score (2019)



Source: Heritage Foundation, Economic Freedom Index 2019

```
# Create Trade Freedom categories (quintiles)
economic_freedom_data <- economic_freedom_data %>%
  mutate(TradeFreedom_Category = cut(Trade.Freedom,
                                     breaks = quantile(Trade.Freedom, probs = seq(0, 1, 0.2), na.rm = TRUE),
                                     include.lowest = TRUE,
                                     labels = c("Low Trade Freedom",
                                                "Lower-Middle Trade Freedom",
                                                "Upper-Middle Trade Freedom",
                                                "High Trade Freedom",
                                                "Very High Trade Freedom"))))

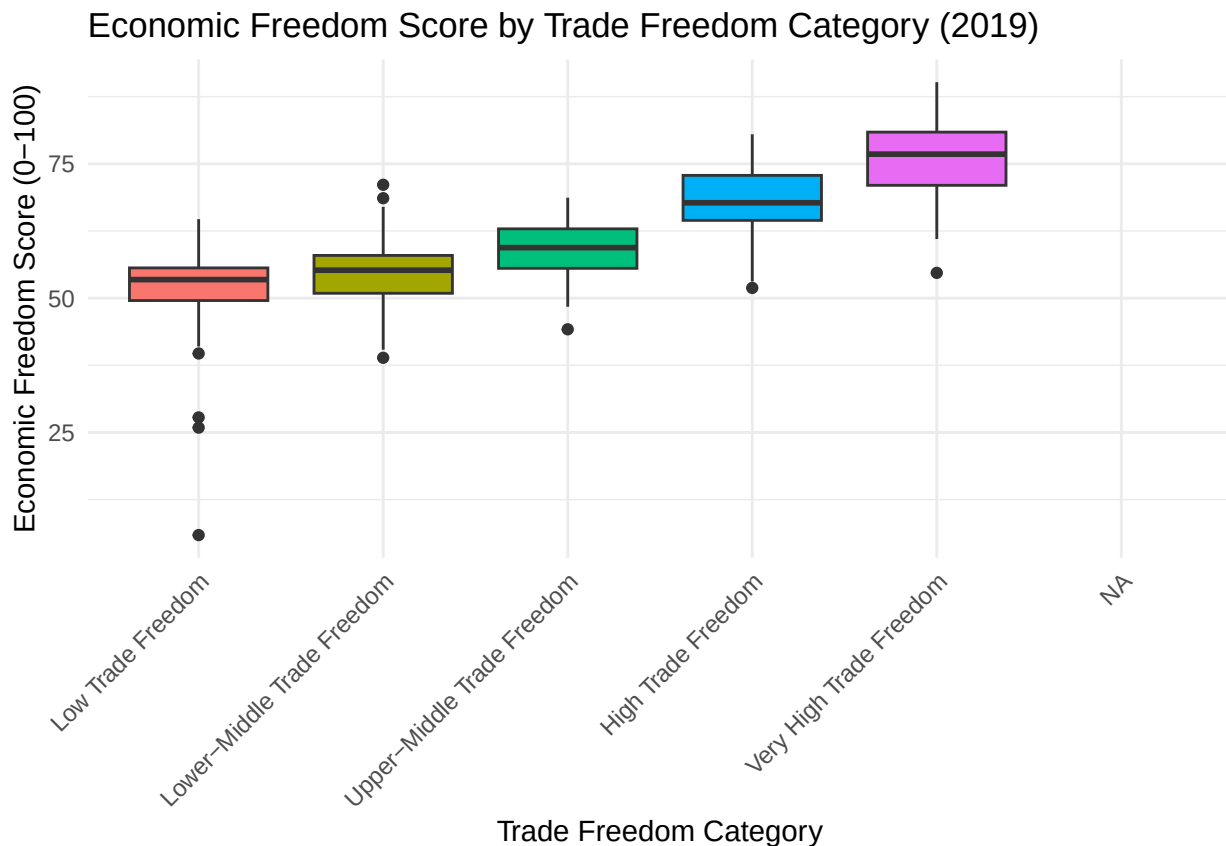
# Summary of Economic Freedom by Trade Freedom category
trade_freedom_summary <- economic_freedom_data %>%
  group_by(TradeFreedom_Category) %>%
  summarise(
    Mean_Freedom = mean(X2019.Score, na.rm = TRUE),
    SD_Freedom   = sd(X2019.Score, na.rm = TRUE),
    Min_Freedom  = min(X2019.Score, na.rm = TRUE),
    Max_Freedom  = max(X2019.Score, na.rm = TRUE)
  )

# Display the summary
```

```
trade_freedom_summary
```

```
## # A tibble: 6 x 5
##   TradeFreedom_Category    Mean_Freedom SD_Freedom Min_Freedom Max_Freedom
##   <fct>                <dbl>      <dbl>      <dbl>      <dbl>
## 1 Low Trade Freedom      50.6       11.1        5.9       64.7
## 2 Lower-Middle Trade Freedom 54.7        7.50       38.9       71.1
## 3 Upper-Middle Trade Freedom 59.0        5.85       44.2       68.7
## 4 High Trade Freedom     67.8        6.45       51.9       80.5
## 5 Very High Trade Freedom  75.4        9.29       54.7       90.2
## 6 <NA>                  NaN         NA         Inf        -Inf
```

```
# Boxplot of Economic Freedom by Trade Freedom category
ggplot(economic_freedom_data,
  aes(x = TradeFreedom_Category, y = X2019.Score, fill = TradeFreedom_Category)) +
  geom_boxplot() +
  labs(
    title = "Economic Freedom Score by Trade Freedom Category (2019)",
    x = "Trade Freedom Category",
    y = "Economic Freedom Score (0-100)",
    fill = "Trade Freedom Category"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none")
```



Conclusion

The results of the analysis show a strong positive correlation between trade freedom and economic freedom. The scatterplot demonstrates that countries with greater trade openness tend to achieve higher levels of economic freedom, highlighting the importance of free trade policies and reduced tariff barriers. The boxplot further supports this relationship, showing that nations in higher trade freedom categories consistently exhibit higher average economic freedom scores.

Are there regional differences in economic freedom scores across the world?

```
# Group data by region and summarize economic freedom scores
```

```
regional_freedom_summary <- economic_freedom_data %>%
  group_by(Region) %>%
  summarise(
    Mean_Freedom = mean(X2019.Score, na.rm = TRUE),
    SD_Freedom   = sd(X2019.Score, na.rm = TRUE),
    Min_Freedom  = min(X2019.Score, na.rm = TRUE),
    Max_Freedom  = max(X2019.Score, na.rm = TRUE)
  ) %>%
  arrange(desc(Mean_Freedom))
```

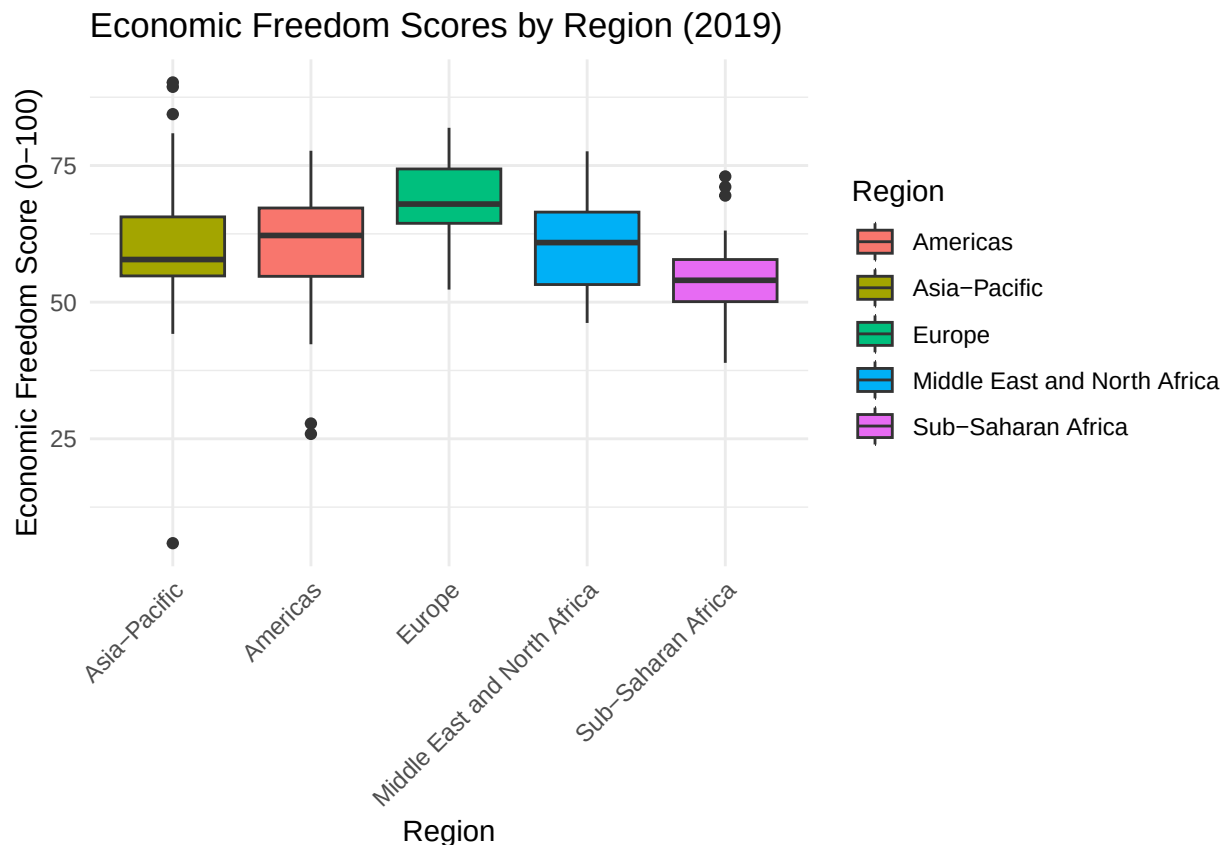
```
# Display the summary
```

```
regional_freedom_summary
```

```
## # A tibble: 5 x 5
##   Region                                Mean_Freedom SD_Freedom Min_Freedom Max_Freedom
##   <chr>                                <dbl>      <dbl>      <dbl>      <dbl>
## 1 Europe                                68.6        6.85       52.3       81.9
## 2 Middle East and North Africa         61.3        9.32       46.2       77.6
## 3 Asia-Pacific                         60.6       13.7        5.9       90.2
## 4 Americas                             59.6       11.9       25.9       77.7
## 5 Sub-Saharan Africa                   54.2        7.22      38.9       73
```

```
# Boxplot of Economic Freedom Scores by Region
```

```
ggplot(economic_freedom_data,
  aes(x = reorder(Region, X2019.Score, FUN = median),
    y = X2019.Score,
    fill = Region)) +
  geom_boxplot() +
  labs(
    title = "Economic Freedom Scores by Region (2019)",
    x = "Region",
    y = "Economic Freedom Score (0-100)",
    fill = "Region"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Conclusion

Regional Variability: There are notable differences in economic freedom across regions. Europe has the highest mean economic freedom score, followed by the Middle East and North Africa, and the Asia-Pacific region. In contrast, Sub-Saharan Africa ranks the lowest, showing limited economic freedom overall.

Regions with High Economic Freedom: Europe stands out with the highest average score (68.6), reflecting strong institutions, open markets, and effective property rights protections. These regions typically benefit from transparent governance and stable regulatory environments that promote private enterprise.

Regions with Low Economic Freedom: Sub-Saharan Africa (mean 54.2) and the Americas (mean 59.6) have lower median economic freedom scores. This may be linked to greater government intervention, weaker property rights enforcement, and political instability that constrain market efficiency.

Variability within Regions: The Asia-Pacific and Americas regions exhibit wider ranges of scores, suggesting greater internal diversity. Some economies are highly open and free, while others face regulatory and institutional challenges that limit freedom.

What patterns or similarities emerge between countries with high and low economic freedom?

```
# Select the top 5 countries with the highest economic freedom
top_5_freedom <- economic_freedom_data %>%
  arrange(desc(X2019.Score)) %>%
  select(Country, X2019.Score, Property.Rights, Government.Integrity, Tax.Burden,
         Business.Freedom, Trade.Freedom, Investment.Freedom, Financial.Freedom) %>%
  head(5)

# Select the bottom 5 countries with the lowest economic freedom
bottom_5_freedom <- economic_freedom_data %>%
  arrange(X2019.Score) %>%
```

```
select(Country, X2019.Score, Property.Rights, Government.Integrity, Tax.Burden,
       Business.Freedom, Trade.Freedom, Investment.Freedom, Financial.Freedom) %>%
head(5)
```

```
# Display both tables
top_5_freedom
```

```
##           Country X2019.Score Property.Rights Government.Integrity Tax.Burden
## 1 Hong Kong SAR      90.2           93.3           83.8           93.1
## 2 Singapore         89.4           97.4           95.1           90.4
## 3 New Zealand       84.4           95.0           96.7           71.0
## 4 Switzerland       81.9           85.3           88.0           70.5
## 5 Australia         80.9           79.1           79.9           62.8
## Business.Freedom Trade.Freedom Investment.Freedom Financial.Freedom
## 1           96.4           95.0           90           90
## 2           90.8           94.8           85           80
## 3           91.0           92.4           80           80
## 4           75.4           87.4           85           90
## 5           88.3           87.6           80           90
```

```
bottom_5_freedom
```

```
##           Country X2019.Score Property.Rights Government.Integrity
## 1 Korea, North      5.9           31.6           24.4
## 2 Venezuela         25.9           7.6           7.9
## 3 Cuba              27.8           31.6           37.7
## 4 Eritrea           38.9           35.5           19.7
## 5 Congo, Republic of 39.7           33.2           25.3
## Tax.Burden Business.Freedom Trade.Freedom Investment.Freedom
## 1           0.0           5.0           0.0           0
## 2           74.7           33.9           60.0           0
## 3           48.8           20.0           64.0           10
## 4           81.4           17.7           69.2           0
## 5           59.5           38.2           56.8           45
## Financial.Freedom
## 1           0
## 2           10
## 3           10
## 4           20
## 5           30
```

```
# Calculate the mean and standard deviation for each factor for both groups
freedom_comparison <- data.frame(
  Factor = c("Property.Rights", "Government.Integrity", "Tax.Burden", "Business.Freedom",
            "Trade.Freedom", "Investment.Freedom", "Financial.Freedom"),
  High_Freedom_Mean = sapply(top_5_freedom[3:9], mean, na.rm = TRUE),
  High_Freedom_SD = sapply(top_5_freedom[3:9], sd, na.rm = TRUE),
  Low_Freedom_Mean = sapply(bottom_5_freedom[3:9], mean, na.rm = TRUE),
  Low_Freedom_SD = sapply(bottom_5_freedom[3:9], sd, na.rm = TRUE)
)
```

```
# Display the summary comparison
freedom_comparison
```

```
##                                     Factor High_Freedom_Mean High_Freedom_SD
```

```
## Property.Rights      Property.Rights      90.02      7.608351
## Government.Integrity Government.Integrity  88.70      7.192009
## Tax.Burden           Tax.Burden           NA        13.389287
## Business.Freedom     Business.Freedom     NA        7.834667
## Trade.Freedom        Trade.Freedom        91.44      3.740053
## Investment.Freedom   Investment.Freedom   NA        4.183300
## Financial.Freedom    Financial.Freedom    NA        5.477226
##                      Low_Freedom_Mean Low_Freedom_SD
## Property.Rights      27.9      11.45993
## Government.Integrity 23.0      10.74988
## Tax.Burden           NA        32.19576
## Business.Freedom     NA        13.33165
## Trade.Freedom        50.0      28.33231
## Investment.Freedom   NA        19.49359
## Financial.Freedom    NA        11.40175
```

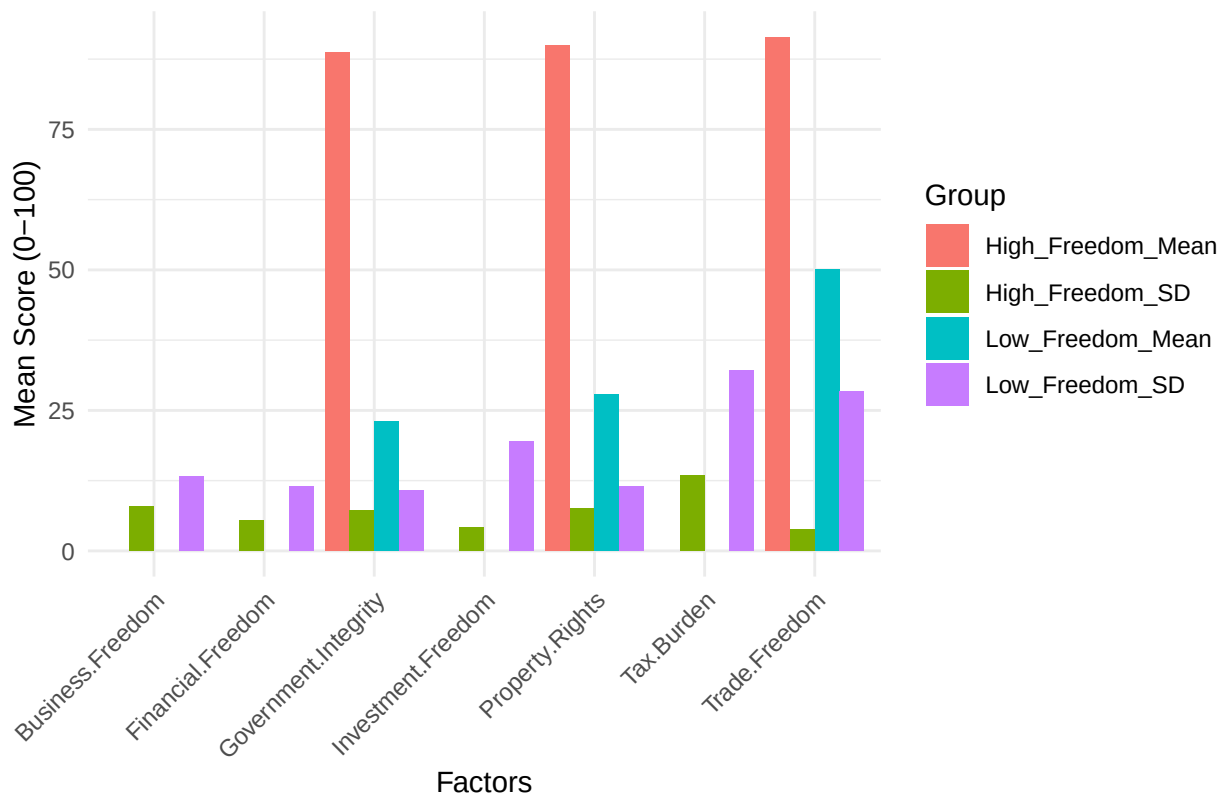
```
# Melt the data for easier plotting
```

```
freedom_comparison_melted <- melt(freedom_comparison, id.vars = "Factor",
                                variable.name = "Group", value.name = "Value")
```

```
# Plot comparison of key factors: High vs Low Economic Freedom
```

```
ggplot(freedom_comparison_melted, aes(x = Factor, y = Value, fill = Group)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Comparison of Key Factors: High vs Low Economic Freedom Countries",
       x = "Factors",
       y = "Mean Score (0-100)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Comparison of Key Factors: High vs Low Economic Freedom Countries



Conclusion

Economic Institutions: Countries with high economic freedom tend to have stronger institutions, reflected in significantly higher property rights and government integrity scores. These nations generally maintain transparent legal systems and low levels of corruption, creating a stable environment for economic activity.

Business and Trade Environment: High-freedom countries exhibit more open markets, with higher trade, investment, and financial freedom scores. These factors suggest that fewer restrictions on capital flows and international trade foster greater overall economic freedom.

Taxation and Regulation: Low-freedom countries typically have higher tax burdens and more restrictive regulatory frameworks, which limit private enterprise and discourage investment.

Overall Patterns: In short, the data indicate that economic openness, institutional integrity, and protection of property rights are key commonalities among high-freedom nations, while excessive government intervention and weak rule of law are consistent traits of low-freedom countries.

Final Policy Implications

The results of this analysis highlight that higher economic freedom is most strongly associated with strong institutions—particularly well-protected property rights, judicial effectiveness, and government integrity. Regions with lower freedom often face structural challenges such as corruption, weak governance, and trade barriers. Policymakers should prioritize institutional reforms that strengthen rule of law, enhance transparency, and promote open markets. Encouraging stable regulatory frameworks and reducing trade restrictions can help countries move toward greater economic freedom and long-term prosperity.